# Downstream-Gated Correction-Linked Memory for Evolving Long-Horizon Dialogues

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**Abstract**

Long-running programming agents must preserve decisions while avoiding constraints that were later corrected, renamed, reverted, or deleted. Static retrieval can surface obsolete statements, whereas deletion alone may discard the correction needed to recover current state. We introduce Correction-Linked Lifecycle Memory (CLLM), a bounded sidecar for an existing memory retriever. Write-time correction events create predecessor–successor links, query-time traversal redirects stale hits, and controlled feedback proposes retrieval policies without retraining the extractor or replacing the base index. We separate proxy optimization from policy promotion: a challenger replaces the incumbent only after predeclared downstream quality, forgetting, cost, and fallback checks pass.

We evaluate the method in MemoryCore on the public Memora long-dialogue benchmark as a methodological proxy for long programming sessions. The fixed-budget V1 policy improves answer-level FAMA from 0.319 to 0.400 on 50 frozen Base-stale-exposed questions, a gain of 8.09 points (95% persona-cluster CI: 5.17–12.59). MiniMax-M3 and DeepSeek-V4-Flash serve as both readers and judges; the primary crossed estimator excludes self-judgment. A larger V2 policy, selected from 36 candidates using direct feedback, improves the direct quarterly proxy over V1 but fails to improve answer FAMA ($-0.43$ points; 95% CI: $-2.06$–1.09), reduces FAA by 4.08 points (95% CI: $-7.62$–$-0.64$), and uses 37.5% more context. The gate therefore retains V1. A post-hoc audit finds more exact obsolete-value exposure in V2 contexts and answers, but does not identify which jointly changed policy dimension caused the regression. Separately, an evidence-preserving compression follow-up on a checksum-frozen 30-question LongMemEval-V2 procedure split holds direct answer-atom support exactly equal to Base on all 12 scored questions while reducing mean injected tokens from 2,796.8 to 2,213.9 ($-20.84\%$). It fails its predeclared *quality-gain* gate because no question improves, but forms a Pareto efficiency result on the measured support–cost plane; no answer-level claim is made. These results show why memory self-optimization should be downstream-gated and why quality and efficiency promotion require distinct objectives. All switch-off and forced-failure tests return the original Base path exactly. The evidence supports a bounded correction-linked mechanism, a safe compression result, and a conservative optimization protocol, not effectiveness on private programming-agent traffic.

## 1 Introduction

Interactive programming tasks evolve. A user may replace REST with GraphQL, rename an interface, invalidate a test expectation, revert an implementation plan, or switch environments. A memory system that retrieves both the old and new statements can confidently reintroduce a rejected constraint. Permanently deleting the old record, however, loses provenance and may prevent the system from finding the correction that explains current state.

Long-term agent memory research studies hierarchical context management [5], dynamic organization [11], scalable extraction and consolidation [1], and temporal knowledge graphs [6]. Benchmarks such as LoCoMo [3] and LongMemEval [10] reveal failures in temporal reasoning and knowledge updates. Memora makes forgetting explicit through invalidated-memory annotations and Forgetting-Aware Memory Accuracy (FAMA) [9]. A central deployment gap remains: a policy can improve retrieval-side evidence coverage while degrading the answer that consumes that evidence.

We ask four questions. **RQ1:** can bounded correction links improve stale-exposed answers without increasing the injection budget? **RQ2:** does a richer policy selected by direct memory feedback also dominate after answer generation? **RQ3:** can a predeclared promotion rule detect and reject a proxy-selected regression? **RQ4:** can an evidence-preserving challenger reduce injection cost without lowering measured support?

These questions separate mechanism efficacy, quality optimization, efficiency optimization, and deployment safety.

We study these questions in MemoryCore, an open-source layered memory service with L0 conversation retrieval and higher-level memories [8]. Our contributions are:

- a replaceable write-event interface and bounded lifecycle ledger above the existing FTS/vector retriever;
- fixed-budget correction-linked redirection that preserves the latest correction rather than deleting provenance;
- a bounded contextual policy search over confidence, traversal, query intent, and injection budget;
- proposal–promotion separation that retains the incumbent when a proxy-selected challenger fails downstream quality, forgetting, or cost checks;
- a checksum-frozen evidence-preserving compression study that reports a 20.84% token reduction separately from answer-quality claims;
- a protocol-versioned public evaluation with direct evidence, a two-reader/two-judge crossed panel, exact fallback checks, an independently recomputed result, and an explicit negative finding.

# 2 Related Work

MemGPT treats context as a hierarchy managed like virtual memory [5]. Mem0 dynamically extracts, consolidates, and retrieves user information, and A-MEM organizes notes into an evolving linked network [1, 11]. Zep/Graphiti represents changing facts in a temporally aware graph [6]. Our scope is narrower: we do not replace the store or build a general knowledge graph. We add bounded invalidation and successor edges immediately above an existing candidate list.

LoCoMo evaluates very long multi-session dialogue [3]; LongMemEval isolates extraction, multi-session reasoning, temporal reasoning, updates, and abstention [10]. Memora is particularly suitable here because its public protocol separates memory-presence from forgetting-absence criteria [9]. We use it for method validation, not as evidence about private programming interactions.

Feedback-driven agents such as Reflexion update behavior from verbal or scalar feedback without weight training [7]. Memory-R1 learns structured add, update, delete, and no-op operations together with answer-time memory use [12]; SelfMem lets an agent refine its memory strategy from environmental feedback [13]; and DeferMem trains query-time evidence selection and rewriting with gated rewards [14]. These systems establish that memory behavior can be learned rather than fixed. Our scope is complementary and deliberately narrower: we evolve a bounded sidecar policy for an existing retriever, preserve a hard Base fallback, and study whether the proxy used to propose a policy agrees with downstream forgetting-aware answers. The observed rank reversal motivates explicit promotion safety rather than a single optimization score.

# 3 Method

## 3.1 Correction-linked retrieval

Let the base retriever return a ranked candidate pool $B(q) = (m_1, \ldots, m_K)$. A write-time adapter emits correction events

$$e = (t_e, c_e, V_e^-, S_e, s_e),$$

where $t_e$ is a monotonic sequence, $c_e \in [0, 1]$ is confidence, $V_e^-$ contains invalidated values, $S_e$ contains successor memory identifiers, and $s_e$ records provenance. A predecessor $m$ receives an edge to $S_e$ when it predates $e$ and contains a value in $V_e^-$.

At query time, the method examines candidates in base rank order. If no eligible edge meets confidence threshold $\tau$, the candidate is unchanged. Otherwise, only the latest eligible correction event is followed. Traversal stops at a leaf, tombstone, hop budget $H$, expansion budget $E$, result budget $k$, or wall-time budget $T$. Duplicate leaves are removed and later base candidates refill the list. Cycles, missing successors, timeouts, malformed state, and capacity violations return $B(q)_{1:k}$ for the entire call.

The research incumbent V1 is $\theta_1 = (\tau = .85, H = 1, E = 64, k = 5, T = 10 \text{ ms})$. It was selected by the original weekly/monthly optimization protocol before quarterly evaluation. The ledger links predecessors directly to the latest eligible correction, so one query-time hop can represent a longer update history.

### 3.2 Contextual challenger

Error analysis motivated two extensions. First, historical aggregates such as totals or week comparisons should not blindly rewrite old evidence into its latest value. A query-text intent adapter can therefore bypass lifecycle traversal for protected aggregate queries. Second, a correction may introduce useful successor evidence outside the original Top-5. We define a dynamic target

$$k'(q) = 5 + \min(s, r(q)),$$

where $r(q)$ is the number of redirects and $s \in \{0, 1, 2\}$ is the extra-slot cap.

The V2 candidate grid contains 36 policies:

$$\tau \in \{.85, .91, .96\}, \quad H \in \{1, 2\}, \quad s \in \{0, 1, 2\}, \quad p \in \{0, 1\},$$

with $E = 64$, $T = 10$ ms, and $p$ controlling aggregate protection. On 300 weekly/monthly questions, utility is

$$U(\theta) = Q_f(\theta) - H_n(\theta) - 0.05C(\theta) - F(\theta),$$

where $Q_f$ is mean direct FAMA proxy on 188 forgetting-bearing questions, $H_n$ is mean positive loss on 112 non-forgetting questions, $C$ is positive extra injected kilotokens, and $F$ is fallback rate. Capacity is fixed at 64 candidate policies. Ties prefer lower harm, cost, hop count, and then higher confidence.

The optimizer selects $\tau = .96$, $H = 2$, $s = 2$, and aggregate protection. This is a proposal, not an automatic deployment decision.

### 3.3 Tiered promotion

We separate policy fitting from promotion. Stage one admits a proposal only if direct quality, protected-slice harm, cost, disabled-equivalence, and forced-failure checks pass. Stage two evaluates the surviving challenger after answer generation. Given incumbent $\theta_I$, challenger $\theta_C$, and predeclared downstream checks $g_1, \ldots, g_J$, the deployed research policy is

$$\theta^* = \begin{cases} \theta_C, & \text{if } \bigwedge_{j=1}^{J} g_j = 1, \\ \theta_I, & \text{otherwise.} \end{cases}$$

The downstream checks cover answer FAMA magnitude and confidence, FAA, relative improvement over the incumbent, and injection cost; stage-one equivalence and fallback checks remain prerequisites. The decision log records every observed value, threshold, and failed check. Thus optimization may learn from feedback without allowing a retrieval-only proxy to silently replace a safer incumbent.

Promotion objectives must also be typed. A *quality* challenger must demonstrate positive downstream utility, whereas an *efficiency* challenger may preserve a predeclared quality invariant while reducing cost by a required margin. These labels cannot be changed after scoring. The D11 follow-up below was frozen with a quality-gain gate and therefore remains rejected for quality promotion; we report its token reduction as a secondary efficiency finding rather than retroactively changing its decision.

## 4 Experimental Protocol

**System and data.** We use MemoryCore L0 SQLite FTS5 retrieval with a candidate pool of 30 and a Base injection limit of five. Memora revision `a6493188efc836d6511ed5e4163fe3ba87da30ff` contains 600 questions, 27,614 sessions, 10 personas, and weekly, monthly, and quarterly horizons. The aggregate data-manifest SHA-256 is `dfc82711...331bfc`. Only turns marked `share_memory` are indexed.

**Data-use boundary.** The ledger uses released write-time fields (session type, operation, operation details, and shared turns), never evaluation evidence. V1 originally optimized on weekly/monthly and held out quarterly. Because quarterly results were observed during V1 development, the V2 protocol correctly treats quarterly as confirmation rather than a pristine new holdout. The 50 answer-level cases are frozen before V2 answers are generated, but they too were used in the earlier V1 study. No external-generalization claim is made.

Memora has no repository, branch, environment, or component scope labels. It can validate correction chains and forgetting, but not programming-specific coexistence. The query-intent classifier uses query text only and matches the benchmark's repeated templates with zero false positives and zero false negatives over 600 questions; this is template validation, not general-language accuracy.

**Protocol chronology.** Protocol v2.0 fixed the candidate grid, objective, split, uncertainty method, and gates before direct scores were computed. A pre-score intent check then found 30 false negatives in two repeated reasoning templates (weekly goal-status and monthly week-comparison questions). Protocol v2.1 changed only those query-text patterns and reran all 600 cases in a fresh output directory; no score, threshold, or candidate dimension was changed. The answer-panel protocol and hashes were recorded before model calls. These are versioned repository artifacts, not a third-party timestamped preregistration, so we use *predeclared* rather than *preregistered*.

**Metrics.** Memory Presence Accuracy (MPA) measures required current facts; Forgetting Absence Accuracy (FAA) measures avoidance of invalid facts. Following Memora,

$$\mathrm{FAMA} = \max(0, \mathrm{MPA} - \lambda(1 - \mathrm{FAA})), \qquad \lambda = \frac{N_f}{N_p + N_f}.$$

The direct proxy counts a current atom only when its annotated source session and normalized value are both retrieved; it counts an obsolete atom only when the value occurs in a candidate predating its invalidation sequence. It applies the same FAMA form to these two rates before answer generation. This provenance-sensitive exact matching is cheap and deterministic, but it cannot score paraphrases, answer wording, or downstream use of a correction statement that repeats an obsolete value. Batched answer-level criteria are not directly comparable to Memora Table 3. All primary confidence intervals use 5,000-sample paired persona-cluster bootstrap.

**Frozen answer panel.** Among 192 quarterly forgetting-bearing questions, 61 have at least one obsolete Base Top-5 hit. Hash stratification selects five cases per persona, producing 50 cases (43 recommending and seven remembering); the selection SHA-256 is `1e3ab7b2...dadaf1`. The V2 context-manifest SHA-256 is `a272a6c5...85b19`. Candidate lists differ from V1 on all 50 cases, so every Base, V1, and V2 answer is generated afresh.

MiniMax-M3 and DeepSeek-V4-Flash [4, 2] each act as reader and judge with thinking disabled. There are $50 \times 3 \times 2 = 300$ reader-arm evaluations and 600 judge calls. The primary crossed estimator uses DeepSeek judgments for MiniMax answers and MiniMax judgments for DeepSeek answers, averaging the two non-self cells per case. A full-factorial sensitivity analysis averages all four cells. Protocols fix exact model IDs, official endpoints, sampling, hashes, gates, and aggregation before calls; no OpenRouter call is used in this final experiment.

**D11 efficiency follow-up.** We additionally evaluate semantic-leaf evidence-preserving substitution on LongMemEval-V2 revision `2cc8c540`. The population contains 106 text-only procedure and procedure-absence questions: 76 are consumed by representation development and audit, and 30 form a checksum-frozen test split, of which 12 have directly computable answer-atom support labels. The all-or-nothing challenger replaces only selected same-trajectory Base items with a bounded capsule containing their semantic values, source provenance, source order, and locally verified actions. Coverage, order, provenance, unrelated-Base preservation, item count, and token certificates must all pass; otherwise the call returns exact Base. Token cost is measured on all 30 questions, while support comparisons use only the 12 labeled questions. The quality gate, artifacts, and admission decision were committed before the test was read. Because that gate requires at least one improved direct-support case, answer generation is not admitted when support is merely equal.

# 5 Results

## 5.1 Direct evidence favors the larger challenger

Table 1 shows the quarterly direct result. V2 exceeds both Base and V1, including on the 61 Base-stale-exposed questions. Its direct gate passes all quality, harm, cost, intent-equivalence, disabled-equivalence, and forced-fallback checks. On all 300 quarterly questions, mean injected tokens rise from 133.68 for Base to 156.76 for V2 (+17.27%).

| Slice | $n$ | Base | V1 | V2 | V2 −Base [95% CI] | V2 −V1 [95% CI] |
|---|---|---|---|---|---|---|
| All quarterly | 300 | 0.423 | 0.439 | 0.445 | +0.022 [0.011, 0.033] | +0.005 [0.002, 0.009] |
| Forgetting-bearing | 192 | 0.450 | 0.477 | 0.481 | +0.031 [0.015, 0.049] | +0.004 [0.001, 0.009] |
| Base-stale-exposed | 61 | 0.091 | 0.184 | 0.194 | +0.103 [0.062, 0.144] | +0.010 [0.003, 0.019] |

Table 1: Direct evidence FAMA proxy on quarterly Memora. Quarterly is a confirmation split for V2, not a pristine external holdout.

For the 100 quarterly reasoning questions, aggregate protection makes V2 exactly equal to Base (FAMA proxy 0.38285), repairing V1's small loss (0.38252). Direct evidence alone would therefore promote the challenger.

## 5.2 Answer-level evidence retains the incumbent

The answer-level result reverses the direct ranking (Table 2). Both lifecycle policies improve substantially over Base. V1, however, has the highest FAMA and FAA. Relative to V1, V2 gains 1.19 MPA points but loses 4.08 FAA points; its FAMA difference is −0.43 points with an interval spanning zero. V2 uses 201.84 injected tokens per case versus 146.76 for V1 (+37.53%).

| Metric | Base | V1 | V2 | V1 −Base [95% CI] | V2 −Base [95% CI] | V2 −V1 [95% CI] |
|---|---|---|---|---|---|---|
| MPA | 0.386 | 0.434 | 0.446 | +0.049 [0.020, 0.093] | +0.061 [0.022, 0.109] | +0.012 [−0.004, 0.028] |
| FAA | 0.853 | 0.937 | 0.896 | +0.084 [0.060, 0.107] | +0.043 [0.003, 0.085] | −0.041 [−0.076, −0.006] |
| FAMA | 0.319 | **0.400** | 0.396 | **+0.081 [0.052, 0.126]** | +0.077 [0.044, 0.114] | −0.004 [−0.021, 0.011] |
| Criterion acc. | 0.704 | **0.761** | 0.738 | +0.057 [0.039, 0.076] | +0.034 [0.012, 0.060] | −0.022 [−0.042, −0.004] |

Table 2: Primary crossed answer-level results on 50 frozen Base-stale-exposed cases.

The full-factorial sensitivity ordering is the same: V1 versus Base is +8.26 FAMA points, V2 versus Base is +7.66, and V2 versus V1 is −0.60. V1 improves FAMA in all 10 persona clusters; the median cluster effect is +5.91 points, and leave-one-persona-out means range from +6.07 to +8.63 points. V2 improves over Base in nine clusters but exceeds V1 in only three.

Figure 1 places the within-stage V2-minus-V1 effects on one percentage-point axis. The direct proxy has a positive interval, while answer FAMA does not improve and answer FAA is significantly negative. Absolute proxy and answer scores are not compared; the figure displays only paired policy differences within each measurement stage.

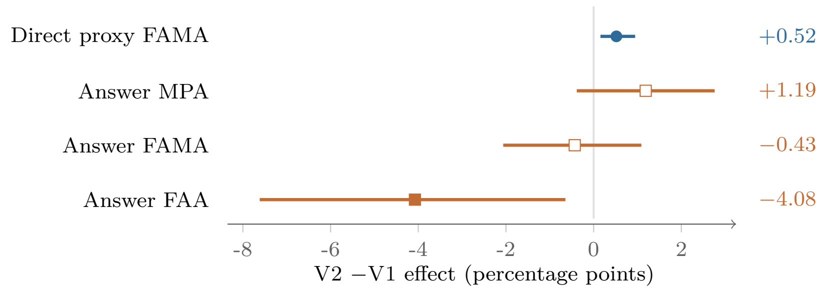


Figure 1: Proxy-to-answer rank reversal. Points are mean V2-minus-V1 effects and lines are 95% persona-cluster bootstrap intervals. Blue denotes the direct proxy; orange denotes answer metrics.

The judges agree on 8,258 of 8,532 paired criterion votes (96.79%, Cohen's $\kappa = 0.899$). The run contains 300 unique reader-arm records, 300 reader calls, and 600 judge calls with no retry and no returned-model mismatch. Three DeepSeek verdicts are `unclear` and counted as incorrect. An independent implementation recomputes all criterion scores and summary means with zero mismatches. A separate 20,000-sample bootstrap with another seed preserves every key inference.

## 5.3 Promotion outcome and failure analysis

The V2 promotion gate passes Base-relative FAMA magnitude, positive FAMA confidence bound, and token budget. It fails the required +10-point Base-relative FAA magnitude (observed +4.31) and positive V2-versus-V1 FAMA direction (observed −0.43). The machine-readable decision is therefore `retain_incumbent`.

The result exposes a proxy-to-generation gap. We therefore run a post-hoc, descriptive exact-value audit over the 100 paired reader answers. At least one normalized obsolete value occurs in 86 V2 contexts versus 74 V1 contexts, and in 81 V2 answers versus 66 V1 answers. V2 also produces longer answers (126.7 versus 112.0 mean completion tokens). Across 1,866 paired forgetting-criterion comparisons, 96 move from correct under V1 to wrong under V2, while 41 move in the opposite direction (net −55). Case inspection includes correction statements and removed to-do items repeated in negated or historical form.

This audit is consistent with increased exposure, but it is not a causal explanation. Exact substring matching misses paraphrases and does not distinguish negation from affirmation; criteria within an answer are correlated; and V2 jointly changes confidence, hops, intent protection, and extra slots. The supported conclusion is therefore policy-level: the selected challenger overfits the direct proxy. Isolating the responsible component requires one-factor answer-level ablations.

An explicitly post-hoc Safe-Hybrid diagnostic uses Base for historical aggregates and V1 for other queries while keeping $k = 5$. It is byte-identical to V1 on the frozen answer sample and passes all routing and fallback equivalence checks. Across all 600 direct cases, however, its FAMA proxy is 0.419076 versus 0.419123 for V1, and its token count is 0.36% higher. It fails a strict no-regression/no-extra-cost diagnostic and is not promoted.

The earlier fixed GPT-4o-mini-reader panel estimated a V1 FAMA gain of 5.77 points (95% CI: 1.23–11.71). It remains secondary provenance evidence; the crossed experiment is primary because answers are regenerated with the selected current models and require no OpenRouter service.

## 5.4 Evidence-preserving compression reduces injection cost

Table 3 reports the untouched D11 test. The candidate substitutes on 19 of 30 questions (63.3%). Mean injected tokens fall by 582.9, or 20.84%, while every directly scored question retains exactly the same answer-atom support as Base. The paired bootstrap interval is $[0, 0]$ because all 12 per-question support differences are zero. Ordered-sequence support is also unchanged. Selection adds 3.08 ms at p95; end-to-end query p95 is 231.1 ms versus 227.5 ms for Base.

| Metric | Base | D11 | D11−Base |
|---|---|---|---|
| Answer-atom support recall ($n = 12$) | 0.208 | 0.208 | 0.000 $[0, 0]$ |
| Ordered-sequence support ($n = 4$) | 0.250 | 0.250 | 0.000 |
| Mean injected tokens ($n = 30$) | 2,796.8 | 2,213.9 | −582.9 (−20.84%) |
| Mean injected items ($n = 30$) | 3.03 | 2.53 | −0.50 |
| Query latency p95 (ms) | 227.5 | 231.1 | +3.6 |

Table 3: D11 evidence-preserving compression on the untouched LongMemEval-V2 procedure test. Support is a direct representation proxy, not generated-answer accuracy.

There are zero improved and zero harmed direct-support cases, zero per-query token violations, zero certificate violations, and zero forced-fallback mismatches; an independent implementation reproduces the result with no mismatch. Under the original quality objective, D11 correctly fails because it creates no newly reachable answer atom, and consequently no reader or judge call is made. Under an efficiency interpretation, however, the observed point is Pareto-improving on the measured support–token plane. We treat this as a useful cost result, not as proof that generated answers are non-inferior: the quality sample contains only 12 direct-proxy questions, and no answer-level equivalence margin was tested.

### 5.5 Safety behavior

Across all 600 questions, disabling CLLM produces zero candidate-ID mismatches against Base. Forced damaged-state and timeout checks also produce zero mismatches, as do protected-intent equivalence checks where applicable. Each group is bounded to 10,000 units, 5,000 events, and 50,000 edges, with query-time hop, expansion, result-count, and wall-time budgets.

## 6 What the Negative Result Changes

The D11 result refines what counts as optimization. Future protocols should predeclare separate promotion classes. A quality challenger must improve downstream effectiveness without violating forgetting or cost guards. An efficiency challenger may instead require a non-inferior quality lower bound, a minimum token reduction, a negative upper confidence bound for token delta, bounded latency overhead, and exact fallback. This permits a genuinely cheaper policy to advance without pretending that zero quality change is a quality gain. Because D11 used the former gate, confirming it as an efficiency policy still requires a fresh split with the latter gate frozen in advance.

The next confirmatory candidate should not be another joint expansion. We retain V1, keep $k = 5$, and vary one dimension at a time: confidence calibration, a second traversal hop, or one additional slot. A two-fidelity optimizer can use the deterministic proxy only to discard clearly weak candidates, then use a development answer panel to estimate forgetting harm before any promotion decision. The current 50 cases must not be reused to fit that answer-level objective; they are now analysis data.

For each one-factor candidate, the decisive estimand is paired answer FAMA relative to V1 with FAA as a non-inferiority safety outcome and injected tokens as cost. An additional slot should be admitted only if its own ablation clears those checks. External confirmation should then use an untouched public update/abstention split or internal programming sessions with repository, branch, environment, and task scope. This sequence directly tests the failure identified here instead of optimizing the same surrogate more aggressively.

Three questions remain open: whether a cheap high-precision surrogate can predict answer-level forgetting harm, whether promotion decisions are stable under a broader reader/judge panel, and how version chains should represent facts that remain concurrently valid in different branches or environments. Each requires new data or a new frozen split; none can be answered by further tuning on the current 50 cases.

## 7 Limitations and Transfer

First, Memora is personalized dialogue, not a programming-agent benchmark. Internal sessions may contain branch-local truths, implicit corrections, code moves, and tool-grounded changes that do not quote the invalidated value. Second, the answer sample is conditional on Base stale exposure and has 43 recommendation versus seven remembering questions; it is not a population estimate over all 600 questions. Third, quarterly and the frozen sample were observed during earlier V1 development, so the V2 results are confirmation and ablation evidence rather than pristine external validation. Fourth, the two-model crossed panel reduces self-judging bias but is not a random sample of readers or judges. Fifth, the post-hoc exposure audit diagnoses association after seeing the outcome and cannot identify a causal policy component. Sixth, the protocol artifacts are versioned but not externally timestamped preregistrations. Seventh, neither policy clears every strict production-style magnitude gate; V1 is the best-tested research incumbent, not an unconditional default. Eighth, D11's support result covers only 12 direct-proxy questions and no generated answers, so its 20.84% token reduction demonstrates measured compression efficiency rather than answer-level non-inferiority.

For programming transfer, the core ledger, policy bounds, decision log, promotion rule, and fallback remain unchanged. Internal adapters must supply stable unit IDs, monotonic events, correction provenance, calibrated confidence, and query scope such as repository, branch, environment, component, and task. Explicit user corrections, rename/refactor events, reverted plans, changed tests, and tool-confirmed deletion are high-value signals. Concurrently valid branch-specific alternatives should coexist rather than invalidate one another. Those assumptions require an internal adapter and a new task-level protocol; Memora cannot validate them.

# 8 Conclusion

Static retrieval can convert historical truth into current error. Correction-linked redirection is a useful bounded mechanism: the fixed-budget V1 policy substantially improves crossed answer-level FAMA at nearly constant context cost and with exact fallback behavior. A more aggressive proxy-optimized V2 policy looks better before answer generation but loses forgetting safety and fails to beat V1 downstream. D11 adds a complementary result: evidence-preserving substitution reduces mean injection by 20.84% with identical measured support, although its small direct-proxy sample and lack of answer calls prevent an answer-quality claim. The central lesson is therefore procedural as well as algorithmic: memory self-optimization should type its objective, propose bounded challengers, test them against frozen quality or efficiency gates, and retain the incumbent on failure. This warrants an internal programming-session pilot, not a broad deployment claim.

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